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Metrics of Automatic Image Quality Assessment based on Human Perception

A Comparative Study and a Proposal of a New Metric

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Metrics of automatic Image Quality Assessment based on Human Perception — A comparative study and a proposal of a new metric

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Resumo

Mantém-se uma discrepância persistente entre as métricas padrão de Avaliação da Qualidade de Imagem (IQA) e os juízos perceptivos humanos, geralmente quantificados através dos Mean Opinion Scores (MOS). Esta divergência constitui um desafio central em aplicações onde a qualidade percebida afeta o desempenho, como no reconhecimento facial. Apresentamos uma nova métrica de Avaliação da Qualidade de Imagens Faciais (FIQA) sem referência, desenvolvida no âmbito de uma estrutura de aprendizagem do tipo Full-to-No-Reference. O processo tem início com um modelo de fusão com referência completa, treinado para fazer a regressão de métricas IQA clássicas em função dos MOS humanos num subconjunto anotado. Este modelo é então utilizado para gerar pseudo-MOS para o conjunto completo de dados. Estes rótulos supervisionam depois um regressor profundo sem referência baseado em características extraídas da ResNet-18, originando uma métrica alinhada com a perceção humana que estima a qualidade diretamente a partir de imagens faciais degradadas. Testámos a nossa abordagem no contexto de imagens faciais afetadas por esteganografia, demonstrando a sua eficácia em cenários com distorções subtis e anotações humanas limitadas.

Abstract

A persistent gap remains between standard Image Quality Assessment (IQA) metrics and human perceptual judgments, typically quantified via Mean Opinion Scores (MOS). This poses a key challenge in applications where perceived quality impacts performance, such as facial recognition. We introduce a new no-reference Face Image Quality Assessment (FIQA) metric, developed within a Full-to-No-Reference learning framework. The process begins with a full-reference fusion model trained to regress classical IQA scores against human MOS on a labeled subset. This model is used to generate pseudo-MOS scores for the full dataset. These labels then supervise a no-reference deep regressor based on ResNet-18 features, producing a perceptually aligned metric that estimates quality directly from distorted facial images. We tested our approach in the context of steganographically degraded facial images, showing its effectiveness in scenarios involving subtle distortions and limited human annotations.

"O que acontece no Mundo é que toda a gente que nasce, nasce de alguma maneira poeta. Inventor de qualquer coisa que não havia no Mundo ainda, antes deles nascerem. E inteiramente individual. Cada um poeta que é!"

Agostinho da Silva

Contents

List of Acronyms

IQA	Image Quality Assessment
ITU-R BT.500-15	International Telecommunication Union - Radiocommunication Sector Recommendation Broadcasting service (television) 500-15
FR	Full-Reference
NR	No-Reference
RR	Reduced-Reference
FIQA	Facial Image Quality Assessment
ICAO	International Civil Aviation Organization
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission
MRTDs	Machine-Readable Travel Documents
MOS	Mean Opinion Score
DMOS	Difference Mean Opinion Score
SS	Single Stimulus
DS	Double Stimulus
FRLL	Face Research Lab London
HVS	Human Visual System
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error

PSNR	Peak Signal-to-Noise Ratio
PSNR-B	Blocking Effect Aware Peak Signal-to-Noise Ratio
BEF	Blocking Effect Factor
SNR	Signal-to-Noise Ratio
SAM	Spectral Angle Mapper
ERGAS	Error Relative Global Dimensionless Attribute Similarity
SSIM	Structural Similarity Index Metric

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1 Introduction

1.1 Contextualization

Image quality refers to the degree to which a visual representation meets perceptual or functional expectations. It is associated with the presence or absence of distortions, artifacts, or degradations that influence how images are perceived by humans or processed by machines. High-quality images preserve structural, textural, and color information in a way that aligns with human visual preferences and supports reliable computer vision performance [1], [2].

Image Quality Assessment (IQA) is the process of quantifying image quality in a systematic and reproducible manner. It plays a central role in applications such as image acquisition [3], denoising [4], compression [5], [6], transmission [7], [8], and restoration [9]. Surveys and comparative studies [1], [10], [11], [12] highlight the diversity of approaches, ranging from signal-based models to data-driven learning techniques, reflecting the complexity of modeling human visual perception.

IQA methods are generally divided into subjective and objective categories. Subjective assessment relies on human observers, typically following standardized protocols such as International Telecommunication Union - Radiocommunication Sector Recommendation Broadcasting service (television) 500-15 (ITU-R BT.500-15) [13], and provides the most direct alignment with perceptual quality. However, it is costly, time-consuming, and not scalable. Objective methods, by contrast, employ computational models to automatically estimate quality, offering scalability and repeatability. Their effectiveness depends on how well they approximate subjective judgments, a long-standing challenge documented in IQA benchmarks and reviews [10].

Within objective IQA, three main categories are distinguished: Full-Reference (FR) [1], No-Reference (NR) [14], and Reduced-Reference (RR) [15]. FR approaches assume access to pristine reference images, RR methods use partial reference information, and NR approaches attempt to predict quality without any reference, making them the most challenging and increasingly important for real-world scenarios.

A specialized subfield, Facial Image Quality Assessment (FIQA), addresses the assessment of face images with respect to both perceptual fidelity and biometric utility [16]. High-quality facial images are essential for accurate recognition in security, forensics, and surveillance. Unlike general IQA, FIQA must account for unique challenges such as pose, occlusion, and demographic variability [17], [18]. This reflects a broader trend in IQA research toward domain-specific assessment strategies [12].

The International Civil Aviation Organization (ICAO) [19] and the International Organization for Standardization / International Electrotechnical Commission (ISO/IEC) 19794–5 standard [20] establish guidelines for image quality in Machine-Readable Travel Documents (MRTDs). These guidelines ensure uniform image conditions (e.g., lighting, focus, and resolution) and consistency across datasets. While these regulations establish a technical baseline, they do not account for perceptual biases and demographic variability in FIQA.

FIQA systems have been shown to produce different quality scores depending on a person's age, gender, or skin tone [21], [22]. These differences often happen because the training data includes more examples from certain groups and fewer from others. For example, images of people with darker skin or non-frontal poses are often less common in training sets, which leads to lower quality scores for those individuals [23]. This can cause face recognition systems to perform worse for some groups than others, raising serious concerns about fairness.

The ethical implications of these biases are profound. Political regulations, such as the European Convention on Human Rights (Article 14) [24], the Universal Declaration of Human Rights (Article 7) [25], the General Data Protection Regulation (Article 22) [26], and emerging AI governance frameworks, such as the European Artificial Intelligence Act (2024) [27] and proposals in the Artificial Intelligence Civil Rights Act (2024) of the U.S. Congress [28], aim to prevent discriminatory decisions. Despite these efforts, biases persist, often introduced through the human observers who evaluate facial images for FIQA algorithms.

Evidence from neuroscience further supports the complexity of facial image perception. The fusiform face area, a specialized region in the human brain, is selectively activated by face stimuli [23], [29]. This biological specialization makes FIQA particularly sensitive to both stimulus features (e.g., age, gender, ethnicity, attractiveness) and the demographic background of the observers.

An even greater challenge arises when evaluating the quality of steganographically distorted facial images. Steganography is the practice of concealing information within digital media, typically by subtly modifying pixel values in a way that is imperceptible to human observers [30]. Although visually unobtrusive, these alterations can degrade biometric features and compromise recognition performance. NR-IQA approaches, while not requiring references, are generally not designed to detect such imperceptible, task-relevant distortions.

This issue is especially relevant for MRTDs, where enhanced security is critical. Incorporating steganographic imagery, such as hidden tamper-evident or authentication data, helps protect against increasingly sophisticated threats to identity documents. The necessity for such measures is underscored by growing criminal activity involving IDs. Reports from the European Border and Coast Guard Agency consistently highlight document fraud as one of the most serious threats at EU external borders [31]. Similarly, Europol identifies fraudulent IDs as a key enabler of organized crime and migrant smuggling [32]. INTERPOL's SLTD database currently contains more than 138 million records of lost or stolen travel documents, illustrating the global scale of the problem [33]. In the United States alone, identity fraud losses reached an estimated $43 billion in 2023 [34].

Steganographically embedding cryptographic signals in passport portraits can therefore serve as a complementary line of defense alongside chip-based authentication. This approach directly addresses known threats such as morphing attacks [35], photo substitution [19], and stolen-document misuse [33], while aligning with ICAO's emphasis on secure and standardized travel document images [19].

1.2 Problem Statement

Despite substantial advances in FIQA, existing methods are not designed to capture subtle but security-critical degradations, particularly those introduced by steganographic embedding. Steganography alters pixel statistics in ways that are imperceptible to human observers yet capable of undermining biometric feature extraction. Conventional FIQA approaches, whether based on handcrafted features [36], supervised learning from quality annotations [37], [38], [39], or ranking-based formulations [40], are typically trained on distortion categories represented in standard datasets (e.g., blur, noise, compression). Consequently, they fail to generalize to covert perturbations that do not visibly degrade an image but can critically impair recognition accuracy.

A second limitation lies in fairness. Empirical studies show that FIQA models often inherit demographic biases from their training data, leading to systematically lower quality scores for specific subgroups [21]. In high-stakes applications such as MRTDs, governed by ICAO and ISO/IEC standards [19], [20], these disparities raise profound ethical and legal concerns. At the same time, evidence from border security agencies highlights that document fraud, morphing attacks, and illicit ID use remain prevalent, underscoring the necessity of robust and equitable quality control mechanisms.

This motivates the development of no-reference FIQA methods explicitly sensitive to both perceptual fidelity and biometric utility in the presence of steganographic distortions. To address this gap, we propose a framework that fuses multiple full-reference metrics into a proxy ground truth aligned with subjective quality judgments. This fused metric supervises the learning of a NR-FIQA model tailored to steganographically modified facial images. By embedding considerations of task relevance, perceptual validity, and demographic fairness, our approach aims to provide a principled bridge between human perception, biometric system requirements, and the operational security demands of MRTDs.

2 Related Work

2.1 Subjective IQA

Subjective image quality assessment is grounded in human visual perception and remains the gold standard for evaluating visual fidelity. Standardized by ITU-R BT.500-15 [13], these methodologies rely on controlled laboratory conditions, calibrated displays, and predefined rating procedures to ensure reproducibility and validity.

2.1.1 Assessment Protocols

Protocols are typically divided into two categories: quality assessment, which captures the overall perceived quality of an image, and impairment assessment, which measures degradation relative to a reference. Two methodological paradigms are most common: the Single Stimulus (SS) method, where each image is rated independently, and the Double Stimulus (DS) method, where reference and test images are evaluated comparatively. SS protocols dominate modern datasets due to their simplicity, efficiency, and reduced observer fatigue.

Rating scales can be numerical or categorical. Numerical scales range from 1 to 5, 1 to 11, or 0 to 100, and are used to express either quality (excellent to bad) or impairment (imperceptible to very annoying). Categorical scales rely on descriptive labels, and are commonly found in ITU protocols. The choice of scale depends on whether the target measure is Mean Opinion Score (MOS) or Difference Mean Opinion Score (DMOS), and on the desired sensitivity of the test.

Each test session consists of multiple presentations. For a given presentation defined by condition j, image or sequence k, and repetition r, observer i provides a discrete score u_{ijkr}. The MOS is computed as:

$$\mathrm{MOS}_{jkr} = \frac{1}{N} \sum_{i=1}^{N} u_{ijkr} \tag{2.1}$$

where N is the number of observers. In double-stimulus protocols, the DMOS is defined as the difference between the MOS of the reference and that of the distorted image:

$$\mathrm{DMOS}_{jkr} = \mathrm{MOS}_{\mathrm{ref},jkr} - \mathrm{MOS}_{\mathrm{dist},jkr} \tag{2.2}$$

These values are often accompanied by confidence intervals to quantify statistical uncertainty and by post-screening to remove inconsistent observers. The specific derivations, including standard deviation, confidence intervals, and the ITU-recommended kurtosis-based rejection procedure, are detailed in Appendix A.

2.1.2 Crowdsourcing and Remote Studies

While laboratory-based studies remain the benchmark for reliability, they are costly, time-consuming, and limited in participant diversity. To overcome these constraints, recent works have turned to crowdsourcing platforms such as Amazon Mechanical Turk, Figure Eight, and Prolific [41], [42]. These frameworks enable the collection of tens of thousands of ratings across heterogeneous populations at a fraction of the cost of laboratory tests.

The advantages of crowdsourcing lie in its scalability, ecological validity, and demographic diversity, which allow researchers to capture quality judgments from a broad spectrum of viewing conditions and cultural backgrounds. This is particularly relevant for applications where image quality intersects with biometric fairness, as demographic variability plays a significant role in subjective perception.

However, uncontrolled environments introduce noise: participants use uncalibrated displays, ratings may be inconsistent, and some responses are malicious or inattentive. To mitigate these issues, robust post-screening strategies are employed, including consistency checks, gold-standard reference images, and statistical filtering [41], [43]. Despite these limitations, crowd-sourced IQA datasets have proven critical for training modern deep learning-based models, as they provide both scale and diversity unavailable in controlled laboratory conditions.

2.1.3 Benchmark Datasets

Several benchmark datasets have been curated to support the development and evaluation of IQA models. Early datasets such as LIVE [44], [45], [46], CSIQ [47], TID2008 [48], and TID2013 [49] follow ITU-R BT.500-15-compliant protocols, providing high-quality MOS or DMOS annotations across a wide variety of synthetic distortion types. These databases remain the most widely used for evaluating traditional full-reference IQA metrics.

The IVC [50] dataset further contributed subjective quality scores for compression-related distortions, while SIQAD [51] targeted screen content images, extending IQA research to non-natural imagery. KADID-10k [52] scaled this approach to 10,000 distorted images with 25 distortion types and multiple intensity levels, enabling large-scale training and fine-grained benchmarking. The Waterloo IAA database [53] emphasized robustness testing through massive-scale synthetic distortions and helped reveal model overfitting issues to specific artifact distributions.

More recent contributions shifted toward naturalistic and crowdsourced assessments. KonIQ-10k [43] introduced 10,073 images collected from the web with authentic distortions and over 1.2 million quality ratings, making it one of the largest ecologically valid IQA resources. SPAQ [54] specifically addressed smartphone photography, collecting 11,125 images and crowdsourced perceptual scores to capture the variability of mobile imaging pipelines. PIPAL [55] targeted perceptual quality in image restoration, including outputs from generative models, and has become a key benchmark for deep learning-based perceptual metrics. Similarly, BAPPS [56] provided a large-scale pairwise comparison dataset for learning perceptual similarity directly from human judgments.

Beyond general-purpose IQA, domain-specific datasets remain critical. Remote sensing benchmarks such as ROSIS [57], AVIRIS [58], and HYDICE [59] provide hyperspectral imagery where distortions affect both perceptual and functional quality, but they lack standardized MOS or DMOS annotations. These datasets illustrate the difficulty of extending IQA methodologies into application-driven contexts.

The Face Research Lab London (FRLL) [60] represents a distinct contribution to IQA benchmarking, focusing on high-quality facial ICAO compliant imagery. It contains 102 unique subjects photographed under controlled studio conditions, with consistent illumination, wearing plain white t-shirts, and neutral, frontal poses. In addition to meeting passport-style requirements, the dataset also includes self-reported age, gender, and ethnicity. Attractiveness ratings (on a 1-7 scale from “much less attractive than average” to “much more attractive than average”) for the neutral front faces, provided by 2513 raters aged 17-90, are also included.

This combination of demographic attributes and attractiveness ratings enables not only research that bridges perceptual quality with subjective aesthetic judgments, but also systematic analysis of demographic bias in human evaluations of facial images. These dual annotations make the dataset particularly suitable for studies involving biometric verification and MRTDs, where both technical quality and fairness considerations are critical.

Unlike general-purpose IQA benchmarks, FRLL does not include traditional synthetic distortions. Instead, its value lies in providing a pristine reference set of demographically diverse facial images, which can be systematically altered or embedded with task-driven degradations such as steganography, compression, or adversarial perturbations. This makes FRLL an essential testbed for application-driven IQA, where the objective extends beyond generic perceptual fidelity to domain-specific measures of utility and fairness.

Together, these databases trace the evolution of IQA benchmarking from controlled laboratory studies with synthetic distortions to large-scale, crowdsourced, and application-specific datasets. This progression not only enabled the development of modern no-reference and deep learning-based IQA models but also revealed persistent challenges in generalization to real-world, domain-critical scenarios.

2.2 Objective IQA

Objective IQA refers to the automatic estimation of visual quality using computational models. Unlike subjective approaches that rely on human ratings, objective IQA enables consistent and scalable evaluations, making it essential for applications such as compression, transmission, restoration, and enhancement [61], [62], [63]. Over the past decades, numerous methods have been proposed, shaped by different assumptions about the Human Visual System (HVS) and the statistical properties of natural images. Surveys and comparative studies [64], [65], [66], [67] summarize this evolution, showing a progression from simple error-based fidelity measures to advanced perceptual and learning-based frameworks. Although no metric fully replicates human judgment, objective IQA provides a reproducible and low-cost alternative that supports large-scale benchmarking and real-time quality monitoring.

A principled way to classify IQA methods is through two dimensions: the computational domain in which comparisons are performed and the spectral representation adopted.

2.2.1 Computational Domains

The computational domain defines the signal space where similarity is quantified.

- Spatial domain: Early metrics such as MSE, RMSE, and PSNR operate directly on pixel intensities by measuring absolute differences or signal-to-noise ratios [61]. Despite their simplicity, they fail to account for perceptual masking and structural consistency.

- Frequency domain: Transform-based methods use the discrete cosine transform (DCT), discrete Fourier transform (DFT), or wavelets to capture energy distributions across spatial frequencies. Examples include IFC [68] and VSNR [69], which exploit properties of the HVS such as contrast sensitivity.

- Gradient and saliency domains: These methods emphasize structural edges and perceptually important regions. The gradient magnitude similarity deviation (GMSD) [70] measures gradient consistency, while saliency-based models [71] weight distortions according to visual attention maps.

- Information-theoretic domain: Some metrics interpret quality as the amount of shared information between reference and distorted images. VIF [72] and related approaches build on mutual information, quantifying how much perceptual information can be extracted reliably from a degraded signal.

- Deep feature domain: More recent approaches leverage convolutional neural networks or vision transformers to learn perceptual embeddings. Metrics such as LPIPS [73] and DISTS [67] correlate strongly with human judgments by comparing features extracted from pre-trained networks.

2.2.2 Spectral Representations

The spectral representation specifies the channel configuration used for quality computation.

- Luminance-based: Many classical approaches, such as SSIM [65], focus solely on the luminance channel (Y). This reflects the dominant role of brightness perception in the HVS and simplifies computation.

- Color-based: Extensions to RGB or alternative color spaces, including YCbCr and CIELab, enable the incorporation of chromatic information. Examples include FSIM and its color extension FSIMc [74], or PQM [75], which attempt to capture hue and saturation degradations alongside luminance.

- Multispectral and hyperspectral: In domains such as remote sensing, quality measures generalize to dozens or even hundreds of bands. Metrics like SAM [76], ERGAS [77], and D_λ [50] ensure spectral consistency across wavelengths and are critical for preserving material properties.

This dual perspective highlights the diversity of IQA methodologies and provides a framework for their categorization. Building on it, Table 2.1 presents the 40 FR metrics and Table **??** summarizes the 4 NR metrics considered in this work. Each metric is characterized by its reference type, year of introduction, category domain, and tested databases, showing how the field has advanced from simple pixel-wise error measures to perceptually and semantically informed quality predictors.

2.2.3 Full-Reference IQA

Table 2.1: Summary of the 40 FR-IQA metrics used

Metric	**Year**	**Category**	**Tested Databases**
MSE [65]	2004	Error-based	LIVE, CSIQ, TID2008, TID2013, KADID-10k
RMSE [65]	2004	Error-based	LIVE, CSIQ, TID2008, TID2013, KADID-10k
MAE [65]	2004	Error-based	LIVE, CSIQ, TID2008, TID2013, KADID-10k
PSNR [65]	2004	Error-based	LIVE, CSIQ, TID2008, TID2013, KADID-10k
PSNR-B [78]	2011	Error-based	LIVE, CSIQ, TID2013
SNR [65]	2004	Error-based	LIVE, CSIQ, TID2008
SAM [76]	1992	Spectral-based	AVIRIS, HYDICE, ROSIS, TID2013
ERGAS [77]	2000	Spectral-based	AVIRIS, ROSIS, TID2013, KADID-10k
D_λ [77]	2000	Spectral-based	AVIRIS, ROSIS, TID2013, KADID-10k
SSIM [65]	2004	Structural Similarity	LIVE, CSIQ, TID2008, TID2013
DSSIM [65]	2004	Structural Similarity	LIVE, CSIQ, TID2008
MS-SSIM [79]	2003	Structural Similarity	LIVE, CSIQ, TID2013, KADID-10k
IW-SSIM [80]	2010	Structural Similarity	LIVE, CSIQ, TID2013
L-SSIM [65]	2004	Structural Similarity	—
UQI [75]	2002	Structural Similarity	—
SCC [81]	1995	Structural Similarity	—
CW-SSIM [82]	2009	Wavelet Similarity	LIVE, CSIQ, TID2013
W-SSIM [83]	2009	Wavelet Similarity	LIVE
ESSIM [84]	2006	Edge Similarity	LIVE
G-SSIM [85]	2006	Gradient Similarity	LIVE, TID2008
R-SSIM [86]	2015	Gradient Similarity	—
GMS [70]	2014	Gradient Similarity	LIVE, CSIQ, TID2008
GMSD [70]	2014	Gradient Similarity	LIVE, CSIQ, TID2008
C-SSIM [87]	2017	Color Similarity	TID2013
FSIM [74]	2011	Feature Similarity	LIVE, CSIQ, TID2008, TID2013, KADID-10k
FSIMc [74]	2011	Feature Similarity	LIVE, CSIQ, TID2008, IVC

(continuation)

Metric	Year	Category	Tested Databases
MS-FSIM [74]	2011	Color Similarity	—
SR-SIM [88]	2013	Feature Similarity	LIVE, CSIQ, TID2008
VSNR [69]	2007	Feature Similarity	LIVE, CSIQ
IFC [68]	2005	Information	LIVE
VIF [72]	2006	Information	LIVE, CSIQ, TID2008
VIFp [72]	2006	Information	LIVE
VIFc [72]	2006	Information	—
LPIPS-Alex [73]	2018	Deep Learning	BAPPS, LIVE, CSIQ, TID2013
LPIPS-Squeeze [73]	2018	Deep Learning	BAPPS, LIVE, CSIQ, TID2013
LPIPS-VGG [73]	2018	Deep Learning	BAPPS, LIVE, CSIQ, TID2013
DISTS [67]	2020	Deep Learning	LIVE, CSIQ, TID2013, KADID-10k
WaDIQaM [89]	2017	Deep Learning	LIVE, TID2013
PSM	—	—	—
SPIQ	—	—	—

2.3 Objective IQA: critical comparison and scope

This section contrasts major metric families used in this thesis and motivates which ones are retained for experiments. Detailed formulas and notations are deferred to Appendix A to keep the narrative focused.

2.3.1 Families of full-reference metrics

We group FR metrics into four families: error-based, similarity-based, information-theoretic, and deep learning feature-based. For each family we summarize principle, strengths, failure modes, and expected behavior on subtle distortions.

Error-based

Principle: pixel-wise deviations. Pros: simple, fast, interpretable. Cons: weak perceptual alignment; insensitive to structural rearrangements; often saturate on subtle distortions. Use in this thesis: baseline correlation analysis and fusion inputs. See Appendix A.1 for formulas.

Similarity-based

Principle: structural or feature resemblance in luminance, contrast, texture, phase. Pros: improved alignment with human judgments; robust to some luminance shifts. Cons: sensitivity to geometric misalignment; some variants overfit to specific artifacts. Use in this thesis: primary contributors to fusion. See Appendix A.2.

Information-theoretic

Principle: mutual information preservation under natural scene statistics models. Pros: principled; captures information loss. Cons: higher complexity; modeling assumptions can misfit faces or non-natural images. Use in this thesis: complementary signals in fusion. See Appendix A.3.

Deep learning feature-based

Principle: distances in pretrained deep feature spaces; structure and texture factors. Pros: high correlation with MOS; captures complex distortions. Cons: data bias risk; less interpretable; backbone choice matters. Use in this thesis: tested as fusion inputs and compared in correlation. See Appendix A.4.

2.3.2 FR metrics considered in this thesis

We report the FR metrics actually computed. Equations are in Appendix A.

Table 2.2: Full-reference IQA metrics considered in this thesis. Formulas in Appendix A.

Metric	Year	Family/Category	Representative databases
MSE	2004	Error-based	LIVE, CSIQ, TID2013
RMSE	2004	Error-based	LIVE, CSIQ, TID2013
MAE	2004	Error-based	LIVE, CSIQ
PSNR	2004	Error-based	LIVE, CSIQ, TID2013, KADID-10k
PSNR-B	2011	Error-based (blocking aware)	LIVE, CSIQ, TID2013
SNR	2004	Error-based	LIVE, CSIQ

(continuation)

Metric	Year	Family/Category	Representative databases
SSIM	2004	Similarity-based	LIVE, CSIQ, TID2008, TID2013
MS-SSIM	2003	Similarity-based (multiscale)	LIVE, CSIQ, TID2013, KADID-10k
IW-SSIM	2010	Similarity-based (info-weighted)	LIVE, CSIQ, TID2013
CW-SSIM	2009	Similarity-based (wavelet phase)	LIVE, CSIQ
W-SSIM	2009	Similarity-based (region-weighted)	LIVE
ESSIM	2006	Similarity-based (edge)	LIVE
G-SSIM	2006	Similarity-based (gradient)	LIVE, TID2008
R-SSIM	2015	Similarity-based (region)	reported in literature
L-SSIM	2004	Similarity-based (local)	reported in literature
UQI	2002	Similarity-based (early model)	reported in literature
SCC	1995	Similarity-based (correlation)	reported in literature
FSIM	2011	Feature similarity (PC+GM)	LIVE, CSIQ, TID2008, TID2013
FSIMc	2011	Feature similarity with color	LIVE, CSIQ, TID2008, IVC
SR-SIM	2013	Saliency-weighted similarity	LIVE, CSIQ, TID2008
VSNR	2007	HVS thresholding, frequency	LIVE, CSIQ
IFC	2005	Information-theoretic	LIVE
VIF	2006	Information-theoretic	LIVE, CSIQ, TID2008

(continuation)

Metric	Year	Family/Category	Representative databases
VIFp	2006	Information-theoretic (pixel)	LIVE
VIFc	2006	Information-theoretic (color)	reported in literature
LPIPS-Alex	2018	Deep features	BAPPS, LIVE, CSIQ, TID2013
LPIPS-Squeeze	2018	Deep features	BAPPS, LIVE, CSIQ, TID2013
LPIPS-VGG	2018	Deep features	BAPPS, LIVE, CSIQ, TID2013
DISTS	2020	Deep features (structure+texture)	LIVE, CSIQ, TID2013, KADID-10k
WaDIQaM	2017	Deep learning FR (siamese)	LIVE, TID2013

2.3.3 No-reference metrics included for comparison

We only list NR metrics that appear later in experiments. Formulas in Appendix A.5.

Table 2.3: No-reference IQA or FIQA baselines included in this thesis.

Metric	Year	Type	Representative datasets
PIQE	2016	Classical NR-IQA	LIVE, CSIQ, TID2013
NIQE	2013	Classical NR-IQA (NSS)	LIVE, CSIQ, TID2013
SER-FIQ	2020	NR-FIQA (stochastic embedding)	face benchmarks
MagFace	2021	FIQA (quality-aware embedding)	face benchmarks

2.3.4 What each family gets right and wrong on subtle distortions

Error-based metrics underreact to imperceptible yet task-relevant changes. Similarity-based metrics improve alignment but can be brittle to alignment issues. Information-theoretic metrics capture information loss but depend on modeling assumptions. Deep feature metrics capture complex changes but inherit bias from pretraining. This motivates metric fusion to combine complementary signals.

2.4 Fusion of FR metrics and learning from pseudo-labels

Prior work has fused heterogeneous metrics and exploited pseudo-labels to overcome limited MOS. We place our pipeline in this line of research and point to Appendix B for model details.

2.4.1 Regression models commonly used for fusion

We briefly list models evaluated in this thesis for fusion: Linear and regularized linear models, kernel SVR, tree ensembles, gradient boosting families, and categorical-boosting approaches. Rationales: linear models for interpretability, SVR for flexible margins on small data, ensembles for non-linear interactions, boosting for strong tabular performance.

2.4.2 Why gradient boosting and why CatBoost here

CatBoost provides ordered boosting and robust regularization on small to medium tabular data, captures non-linearities among metrics, and supports feature importance analysis. Details of CatBoost configuration and training used in this thesis are presented in Methodology, Section **??** and Section **??**. Comparative descriptions of alternative regressors are in Appendix B.

2.5 Steganography and printer-proof approaches

We review steganography with emphasis on methods robust to print-scan pipelines, which is the operational context for MRTDs.

2.5.1 Classical embedding domains

Spatial LSB, transform-domain DCT and DWT, and adaptive schemes. Strengths, detectability, and robustness trade-offs.

2.5.2 Printer-proof deep pipelines

End-to-end learned encoders and decoders trained with differentiable print-scan models and geometric-photometric perturbations.

StegaStamp

Key idea, robustness sources, message capacity, typical artifacts. Relevance to high-resolution face portraits.

CodeFace

Face-specific alignment and identity-preserving losses. Relevance to FIQA and MRTD portraits.

RiemStega

Statistical descriptors on SPD manifolds to preserve global structure. Expected behavior under scanner noise and color drift.

StampOne

Frequency-aware encoding with attention across bands. Robustness to printer color shifts and sensor noise.

2.6 Synthesis and implications for our approach

Takeaways. No single FR metric captures imperceptible but recognition-relevant degradations. Families are complementary but biased in distinct ways. Pseudo-label fusion offers a scalable path to align with MOS. Printer-proof steganography demands quality predictors sensitive to subtle, structured perturbations. These observations motivate the fusion-to-NR strategy adopted in this thesis and lead directly to the pipeline in Chapter C.

A Mathematical definitions of IQA metrics

This appendix collects formulas and notations for all metrics referenced in Chapter 2. Keep Notation formatting under **Notation.** blocks only.

A.1 Error-based metrics

A.2 Similarity-based metrics

A.3 Information-theoretic metrics

A.4 Deep feature-based metrics

A.5 No-reference metrics

B Regression models compared

This appendix summarizes alternative regressors evaluated for fusion and provides reference equations and training details.

B.1 Linear and regularized linear models

B.2 Kernel methods

B.3 Tree ensembles

B.4 Gradient boosting families

C Methodology

C.1 Dataset

Our dataset is derived from the publically available Face Research Lab London [60] (FRLL) set, comprising 102, 1350×1350, full coloured images of frontal, ICAO-compliant adult faces. Self-reported age, gender and ethnicity are included. Attractiveness ratings (on a 1–7 scale from "much less attractiveness than average" to "much more attractive than average") for the faces from 2513 people (ages 17–90) are included as well. The FRLL dataset was selected for its high-fidelity ICAO-compliant images, controlled acquisition conditions, and extensive demographic annotations, which are crucial for analyzing perceptual quality and potential demographic biases in subjective ratings. Each image was encoded using four printer-proof steganographic methods, each applied at nine different intensity levels, $x \mid x = 0.6 + 0.1 \times k, k \in \{0, \ldots, 8\}$, yielding a total of 3,672 distorted images.

The dataset was partitioned to support both supervised learning and independent evaluation. A small, demographically diverse subset of 15 identities, shown in Fig. C.1 was annotated with MOS through subjective tests. This set was split into a training portion (12 identities, Fig. C.1a) used to fit the fusion model, and a disjoint test portion (3 identities, Fig. C.1b) held out for final evaluation. The remaining 87 identities formed the unlabeled set, to which pseudo-MOS were later assigned. Together, the annotated and pseudo-labeled subsets were used to train the NR-IQA model. A summary of these partitions, including the number of subjects and images per subset, is presented in Table C.1.

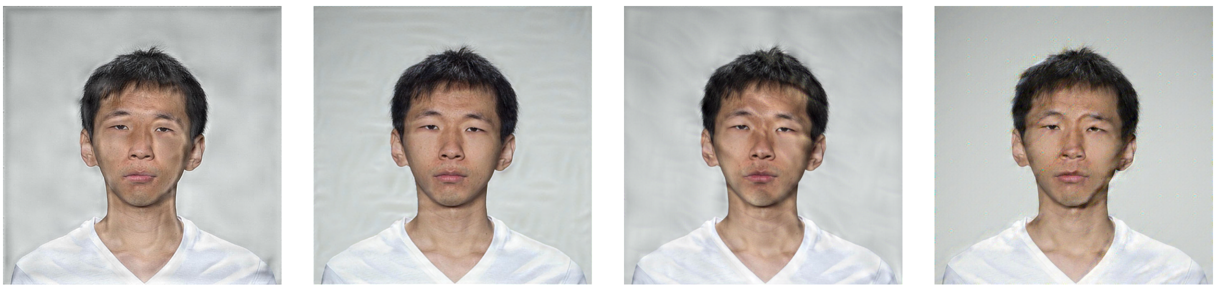
The steganographic distortions are visible in Fig. C.2, which shows examples of distorted facial images from each method. The distortions are applied to the original images, and the resulting stego images are used for both subjective evaluation and training of the NR-IQA model. The selected steganographic methods represent diverse classes of distortion processes (geometric, color-based, local texture perturbation) commonly encountered in practical steganographic applications. The chosen intensity levels ensure a broad perceptual range from barely noticeable to clearly visible distortions.

Table C.1: Partitioning of the steganographically distorted dataset.

Subset	**Subjects**	**Images**	**Purpose**
MOS train set	12	432	Train FR fusion model
MOS test set	3	108	Final evaluation (disjoint)
Pseudo-MOS set	87	3,132	Generate pseudo-MOS labels
NR train set (total)	99	3,564	Train NR model (MOS train + pseudo-MOS)
Total	**102**	**3,672**	Full dataset (all stego levels)

(a) MOS train set (b) MOS test set

Figure C.1: Reference images from the MOS set, comprising the selected subjects from the FRLL dataset.



(a) StegaStamp (b) Code Face (c) RiemStega (d) StampOne

Figure C.2: Steganographically distorted facial images from each method.

C.2 Collection of Subjective Scores

We followed the ITU-R BT.500–15 **ITU-R-BT500** recommendation and adopted the Single Stimulus (SS) method. The test was implemented using a custom Django web application seen in Fig. C.3. Prior to the test session, participants signed an informed consent form and filled out a registration form providing demographic and environmental information such as age, gender, education, country of origin and ethnicity, and others. After this, observers were shown 10 randomly selected sample images. These examples were designed to help them understand the variation in quality across the dataset.

Each image was shown individually, with no time limit. Ratings were submitted using a labeled slider, and automatic saving ensured session robustness.

Each image in the MOS set was evaluated approximately 30 times by human observers, resulting in over 14,000 ratings. We had around 200 participants, each session lasted about 22 minutes and included roughly 70 evaluations. Table C.2 summarizes the demographic profile of the participant pool. Following the session, outlier observers were identified and removed using both Kurtosis-based and correlation-based post-screening methods described in ITU-R BT.500–15 **ITU-R-BT500**, resulting in the exclusion of one participant.

The database was implemented using the Django web framework, which provides an object-relational mapping (ORM) layer that directly maps Python model classes to database tables. The ORM also enforces data integrity constraints and simplifies querying for statistical analysis. The database backend used was PostgreSQL.

The database schema was designed to ensure structured storage and traceability of all subjective quality evaluation data. Fig. C.4a presents the conceptual data model, which defines the main entities and their relationships. The profile entity stores participant metadata, including age, gender, education, ethnicity, country of origin, and device used. The ss_session entity models individual test sessions and is linked to both the participant profile and the dataset being evaluated. The image entity encodes each image's filename, associated distortion type (distortion_name), and distortion level (distortion_level). The test entity stores individual subjective ratings, including the precise timestamp of submission, linked to both the session and the image presented. An auxiliary user_feedback entity captures optional, anonymous, observer comments and critiques.

The physical implementation of this schema, shown in Fig. C.4b, is realized as a relational database with explicit foreign key constraints to ensure referential integrity. The dataset_id and profile_id fields in ss_session formally enforce the linkage to specific datasets and partici-

Figure C.3: Django-based webapp created for the Single Stimulus test.

Table C.2: Demographic and non-demographic profile of the observers.

(a) Age group

Group	Count
18–25	85
26–40	54
41–60	52
Above 60	9
Under 18	2

(b) Gender

Gender	Count
Male	143
Female	59

(c) Ethnicity

Group	Count
White	177
Latino	8
Black	6
Mixed / Multiple	3
Others	7

(d) Education

Level	Count
Bachelor's	92
Master's	63
Doctorate	29
High School	14
Other	4

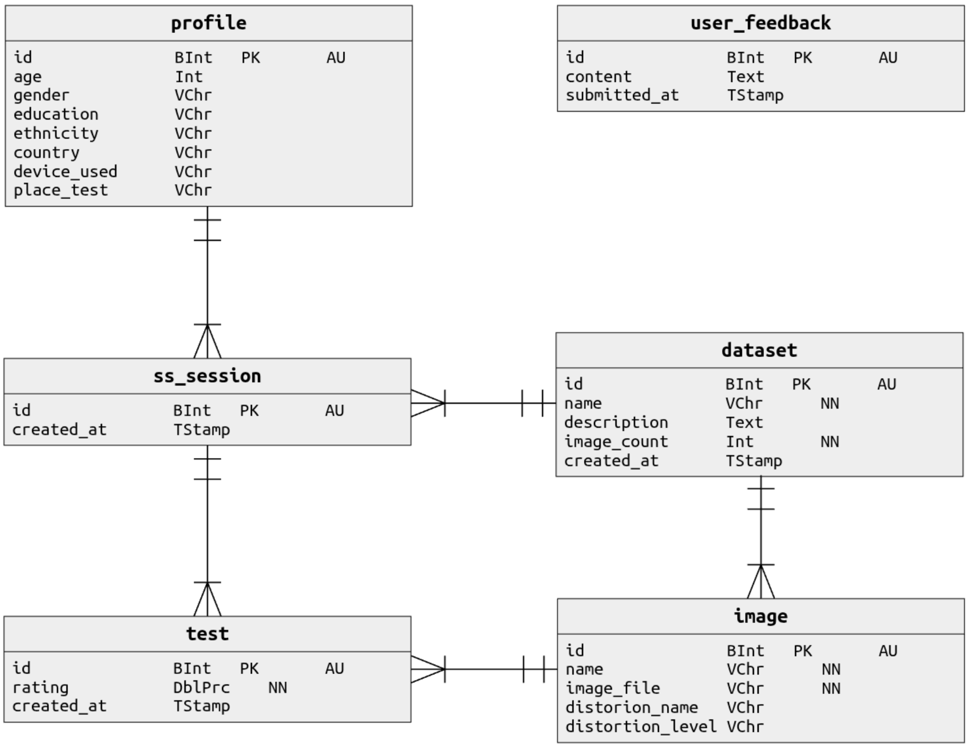
(e) Country

Country	Count
Portugal	170
Brazil	16
Russia	3
India	2
Ghana	2
Others	9

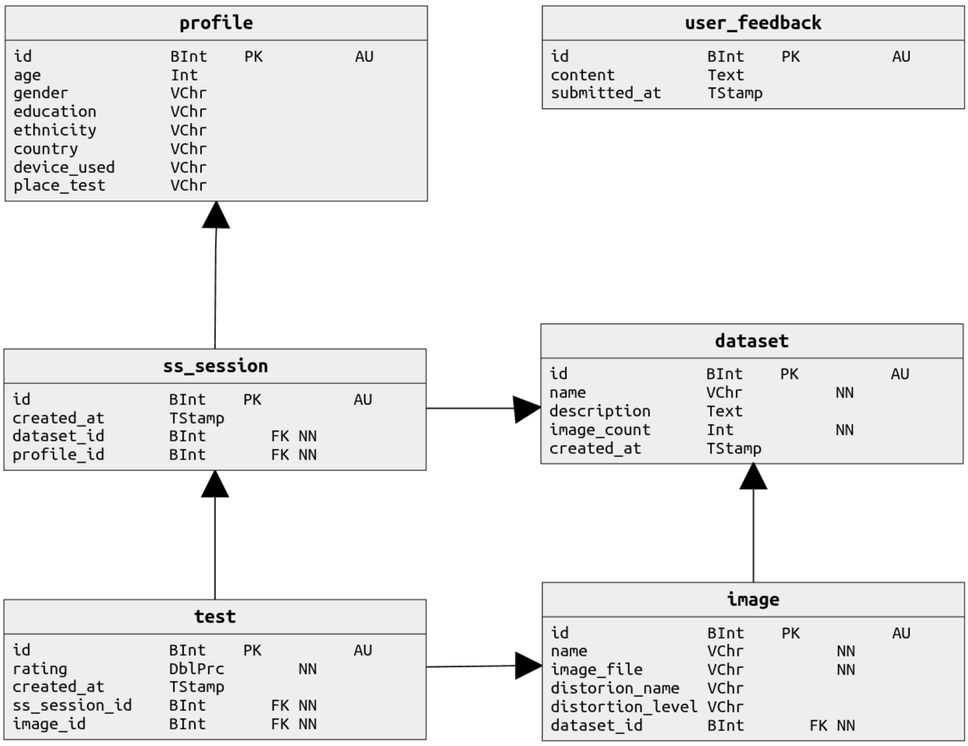
(f) Device Used

Device	Count
Laptop	94
Phone	65
Desktop	42
Other	1

pants. The test table connects each subjective rating to its session (ss_session_id) and image (image_id), enabling consistent aggregation of ratings into a statistical descriptor such as MOS and confidence intervals per image and per distortion level. The relational model also facilitates efficient querying for downstream statistical analysis and supports reproducibility of the experimental protocol in line with ITU-R BT.500-15 recommendations.



(a) Conceptual database schema showing entity relationships and high-level structure.



(b) Physical database schema illustrating actual tables, columns, and implementation details.

Figure C.4: Conceptual and physical representations of the database schema.

C.3 Debiasing of Subjective Scores

To correct for demographic bias in the subjective scores, we followed a procedure inspired by prior work on bias correction in perceptual tasks [90], where we applied a residualization method based on linear modeling. An ordinary least squares (OLS) regression was fit to the MOS values, using observer and image attributes, and their pairwise interactions as categorical predictors. The fitted bias components were subtracted from the original scores, and the residuals were mean-centered to preserve the global score distribution. As shown in Table C.3, several factors exhibit statistically significant effects on the MOS prior to residualization, notably observer and subject ethnicity. After applying the residualization procedure, these effects disappear, as confirmed by an ANOVA test showing no significant impact from any individual factor. The corrected MOS labels are then used as ground truth in all supervised stages of the pipeline to ensure fairness and reduce the influence of socially conditioned priors.

Table C.3: ANOVA [91] results for observer and image attributes. Before debiasing, several factors show statistically significant effects on MOS, p-value < 0.05. After residualization, all main effects show no significant impact, confirming the effectiveness of the debiasing procedure.

Factor	p-value	p-value (residualized)
Observer gender	0.022	0.9930
Observer ethnicity	8.44×10^{-4}	1
Subject gender	1.60×10^{-3}	0.9722
Subject ethnicity	7.36×10^{-3}	1
Observer gender × Subject gender	0.6417	0.6417
Observer ethnicity × Subject ethnicity	0.0582	0.0582

C.4 Correlation of FR-IQA Metrics with Human Perception

We compute 40 FR-IQA scores for each distorted image in the dataset and compare them against the corresponding MOS, as seen in Fig. C.5. Several metrics exhibit strong linear trends with MOS, while others are poorly aligned or even negatively correlated. For a detailed description of these metrics, we refer the reader to [64].

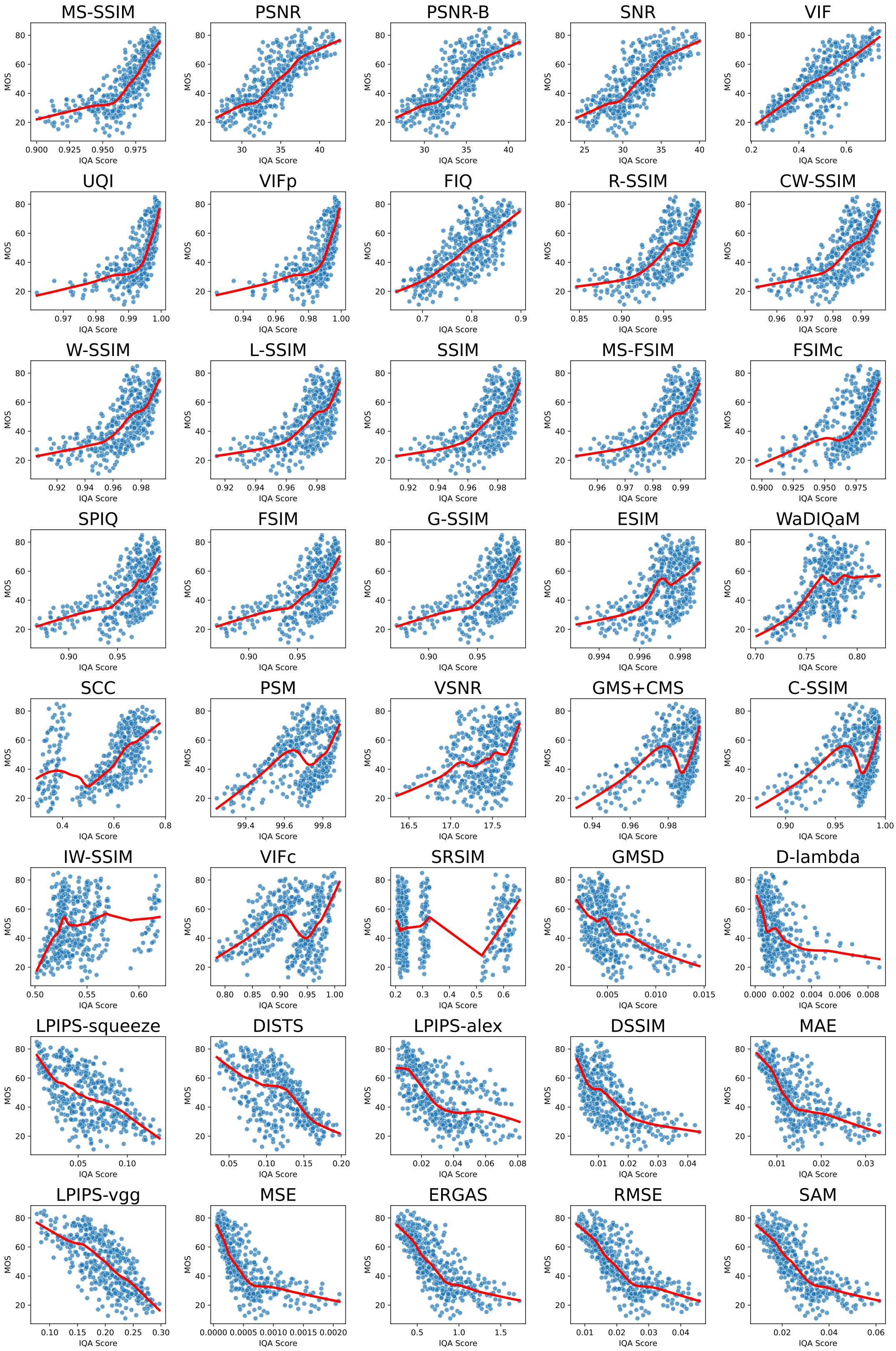


Figure C.5: Scatter plots illustrating the relationship between MOS and 40 individual full-reference IQA metrics. The red line represents a smoothed local trend curve.

C.5 Fusion of FR-IQA Metrics for Pseudo-MOS Estimation

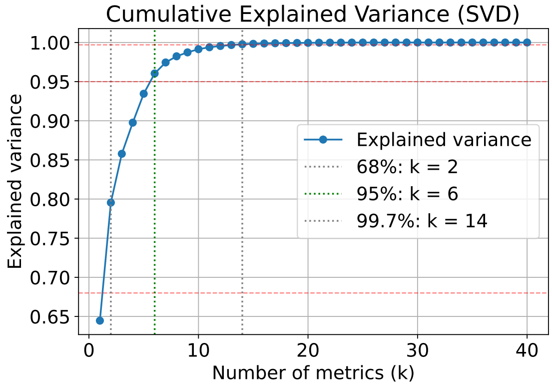
To identify which metrics align best with human perception, we compute both the Pearson Linear Correlation Coefficient (PLCC) and the Spearman Rank-Order Correlation Coefficient (SRCC) [92] which respectively quantify the linearity and monotonicity of the relationship between metric scores and MOS. To determine the appropriate number of metrics to retain for fusion, we applied Singular Value Decomposition (SVD) and the Picard criterion [93]. Metrics are first ranked by the average of their PLCC and SRCC with MOS. After normalizing the feature matrix, SVD revealed that six components capture 95% of the total variance, as shown in Fig. C.6a, indicating an optimal dimensionality of $k = 6$.

To validate this truncation point, we examined the Picard plot in Fig. C.6b, which compares singular values with the target projections. The stable ratio in the tail confirms that six components provide a good balance between expressiveness and stability. This supports a compact, informative subset of FR-IQA metrics.

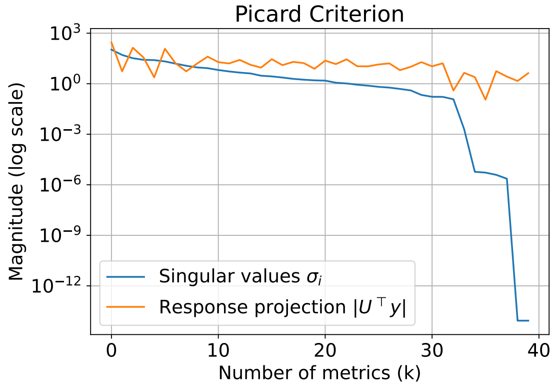
After selecting the top $k = 6$ FR-IQA metrics, we trained a diverse set of supervised regressors to map these features to subjective MOS scores. The models included both linear and non-linear types:

- Linear models: Linear Regression [94], Ridge Regression [95], Bayesian Ridge [96], ElasticNet [97].
- Kernel-based: Support Vector Regression [98] (SVR).
- Ensembles: Random Forest [99], Extra Trees [100], Gradient Boosting [101], HistGradientBoosting [102].
- Boosted Trees: XGBoost[103], LightGBM [104], CatBoost [105].

Each regressor was trained using five-fold cross-validation with an exhaustive grid search over predefined hyperparameter spaces. The best model, CatBoost, optimal configuration was: depth = 8, iterations = 500, learning rate = 0.05, and L_2 regularization = 1. This trained regressor is then applied to the unlabeled portion of the dataset (3,132 distorted images), generating pseudo-MOS.



(a) Cumulative variance explained by SVD components.



(b) Magnitude of singular values and projected response.

Figure C.6: SVD-based analysis of the FR metric space. The cumulative explained variance (a) guides the choice of dimensionality, while the Picard criterion (b) illustrates stability near $k = 6$.

C.6 Training a No-Reference IQA Model from Pseudo-MOS

To enable NR quality prediction, we train a deep regression model end-to-end using pseudo-MOS scores as targets. The architecture is based on a ResNet-18 backbone [106] pretrained on ImageNet [107], with its final classification layer replaced by a lightweight multi-layer perceptron (MLP) regressor [96]. All layers are fine-tuned during training to learn perceptual quality representations specific to our task. The training set consists of distorted facial images paired with either pseudo-MOS, from the FR fusion, or real MOS labels, when available. The model is optimized using MSE and evaluated on the disjoint, MOS test set of labeled images. Once trained, the model infers image quality solely from the distorted input, enabling NR assessment aligned with human perception.

C.7 Results

C.7.1 Full-Reference Fusion Metric Performance

To interpret feature contributions, we used SHAP [108] (SHapley Additive exPlanations) values derived from CatBoost's internal structure. SHAP values quantify how much a feature pushes the model's output away from the expected value. As shown in Fig. C.7, MS-SSIM [79] (Multiscale SSIM) and UQI **wang2004ssim** (Universal Quality Index) had the largest positive impact on predictions, while PSNR-B **ma2011psnr**, VIF**sheikh2006image**, (PSNR-Blocking),

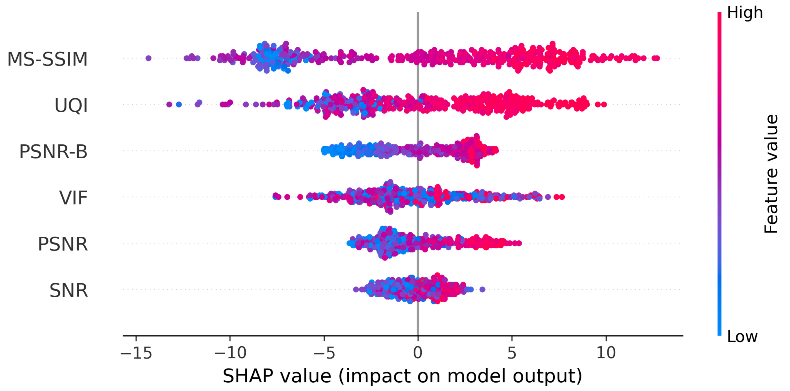


Figure C.7: SHAP summary plot for $k = 6$, generated from the best CatBoost model.

Table C.4: Performance of our Fusion metric and the selected FR-IQA metrics. Higher PLCC and SRCC indicates better correlation with MOS. Lower MSE and MAE indicate more accurate quality estimates.

Metric	**PLCC**	**SRCC**	**MSE**	**MAE**
Fusion (CatBoost)	**0.8881**	**0.8893**	**65.07**	**6.160**
MS-SSIM	0.7712	0.8489	2542	47.27
PSNR	0.8009	0.8132	420.7	16.57
PSNR-B	0.7996	0.8126	434.8	16.84
SNR	0.7931	0.8047	493.9	18.08
VIF	0.7723	0.7671	2584	47.75
UQI	0.6866	0.8085	2540	47.24

PSNR [109] and SNR **gonzalez2002digital** (Signal-to-Noise Ratio) contributed with moderate influence. The color encoding further highlights how high and low feature values affect the model output.

When comparing the fusion metric with the individual FR metrics, Table C.4 shows that the fusion metric outperformed all individual metrics in terms of PLCC, SRCC, MSE and MAE. This demonstrates the effectiveness of our approach in combining multiple FR metrics to improve overall performance.

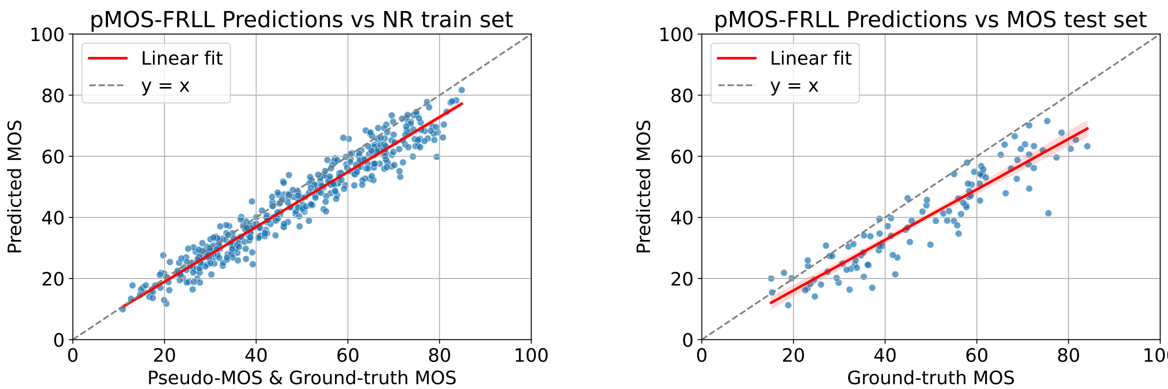
C.7.2 No-Reference Regression Model Performance

We refer to our NR-IQA model as pMOS-Face (pseudo-MOS for Face), reflecting its use of pseudo-MOS labels generated from the FR fusion model and its evaluation on faces. This framework can be adapted to other datasets, provided that a small set of images with human

Table C.5: Performance of our pMOS-Face compared to standard NR-IQA baselines.

Metric	PLCC	SRCC	MSE	MAE
pMOS-Face (ours)	**0.9285**	**0.9345**	**120.6**	**9.158**
NIQE	0.7536	0.7431	6859	82.62
PIQE	0.2993	0.3114	3297	57.05
SER-FIQ	-0.1648	-0.1542	123.5	9.230
MagFace	-0.6095	-0.6362	4437	66.24

labels is available for training the FR fusion model. Fig. C.8 shows two evaluation scenarios: in Fig. C.8(a), the model's predictions are compared against the pseudo and ground-truth MOS used during training, yielding a PLCC of 0.9285, SRCC of 0.9345, MSE of 120.2, and MAE of 9.158. Fig. C.8(b) shows the predictions compared against the MOS test set, resulting in a PLCC of 0.9679, SRCC of 0.9691 and MSE of 35.34 and MAE of 4.743. These results confirm both generalization to unseen human labels and alignment with the pseudo-supervision used for training.



(a) Training plot. **(b)** Test plot.

Figure C.8: Evaluation of our NR-IQA model. Panel (a) shows performance on the training set with pseudo and ground-truth MOS; panel (b) shows performance on the disjoint test set with ground-truth MOS.

The results indicate that our model is more effective in predicting perceptual quality than classical NR-IQA metrics such as NIQE [110] and PIQE [111], as well as task-specific approaches like SER-FIQ [39] and MagFace [38]. This highlights the robustness of our weakly supervised strategy for NR-IQA on steganographically degraded facial images. Quantitative results are reported in Table C.5.

When considering a more generalized use, where observer bias is not accounted for, our NR-

IQA model shows a major improvement over the baselines. This is particularly relevant for task specific applications, where the model can be trained on a small set of images with human labels and then applied to larger datasets without the need for additional supervision. This approach allows for a more scalable and efficient solution not only for FIQA but also for other domains where reference images are not available.

C.8 Conclusion and Future Work

We proposed a Full-to-No-Reference framework for FIQA that predicts image quality in the absence of reference images by leveraging a pseudo-MOS supervision strategy. Our method first trains a FR fusion model to regress human perceptual judgments on a labeled subset (10% of the dataset), generating pseudo-MOS labels for a larger unlabeled dataset. These forged labels are then used to train a deep NR regressor, enabling quality prediction from distorted images alone. This two-stage pipeline effectively bridges the gap between fully supervised FR-IQA and reference-free NR-IQA approaches.

Beyond the development of our NR IQA metric, the proposed framework offers a flexible foundation for constructing a variety of task-specific models. By enabling scalable, perceptually grounded supervision with limited ground-truth annotations, our approach can facilitate quality-aware training in applications such as GANs, forensic imaging, and domain-adapted biometric pipelines.

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Appendix A

Subjective Screening Procedures

This appendix reproduces the statistical screening and post-processing steps recommended by ITU-R BT.500 for subjective image quality assessment (IQA). These procedures were referenced in Chapter 2 but are presented here in full detail for completeness.

A.1 Confidence Intervals

To quantify uncertainty in scores, the sample standard deviation is computed:

$$S_{jkr} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(u_{ijkr} - \bar{u}_{jkr})^2}. \tag{A.1}$$

The 95% confidence interval is then given by:

$$\mathrm{CI}_{95} = 2 \cdot \frac{S_{jkr}}{\sqrt{N}}. \tag{A.2}$$

This interval assumes approximate normality and indicates the range within which the true population mean lies with 95% probability.

A.2 Observer Screening

After score aggregation, observers must be screened to eliminate inconsistent or unreliable participants. ITU-R BT.500 prescribes a kurtosis-based post-screening procedure, especially when the number of subjects is small ($N < 20$) or when non-expert assessors are involved.

For each presentation (j, k, r), the following statistics are computed:

- Mean score $\bar{u}_{jkr}$
- Standard deviation S_{jkr}

- Kurtosis coefficient $\beta_{2,jkr}$, given by:

$$\beta_{2,jkr} = \frac{m_4}{(m_2)^2}, \quad m_x = \frac{1}{N}\sum_{i=1}^{N}(u_{ijkr} - \bar{u}_{jkr})^x \,. \tag{A.3}$$

If $2 \leq \beta_{2,jkr} \leq 4$, the distribution is considered normal; otherwise, it is treated as non-normal. For each observer i, two counters are defined:

- P_i: number of times the observer's score is an outlier above the acceptable range
- Q_i: number of times the observer's score is an outlier below the acceptable range

A.2.1 Outlier Detection

If the distribution is normal:

- If $u_{ijkr} \geq \bar{u}_{jkr} + 2S_{jkr}$, then $P_i = P_i + 1$
- If $u_{ijkr} \leq \bar{u}_{jkr} - 2S_{jkr}$, then $Q_i = Q_i + 1$

If the distribution is non-normal:

- If $u_{ijkr} \geq \bar{u}_{jkr} + \sqrt{20}\, S_{jkr}$, then $P_i = P_i + 1$
- If $u_{ijkr} \leq \bar{u}_{jkr} - \sqrt{20}\, S_{jkr}$, then $Q_i = Q_i + 1$

A.2.2 Rejection Rule

After processing all presentations, observer i is rejected if both conditions hold:

- $\frac{P_i+Q_i}{L} > 0.05$
- $\left|\frac{P_i-Q_i}{P_i+Q_i}\right| < 0.3$

where L is the total number of presentations.

A.3 Summary

The post-screening protocol removes observers whose responses deviate excessively from the group distribution. This procedure improves the reliability of subjective scores and is widely used in IQA studies, particularly when the pool of assessors is small or the distortions under study are subtle.